
Autonomous Production Choke Controller

Example Outputs Guide

Honeywell Hackathon 2026

Engineering Submission

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This folder contains example outputs from the Autonomous Production Choke Controller. All outputs were generated by running `run_scenarios.py` against the TestSimulator at seed=42.

Scenario A: Constant Target (Q=80 bbl/hr)

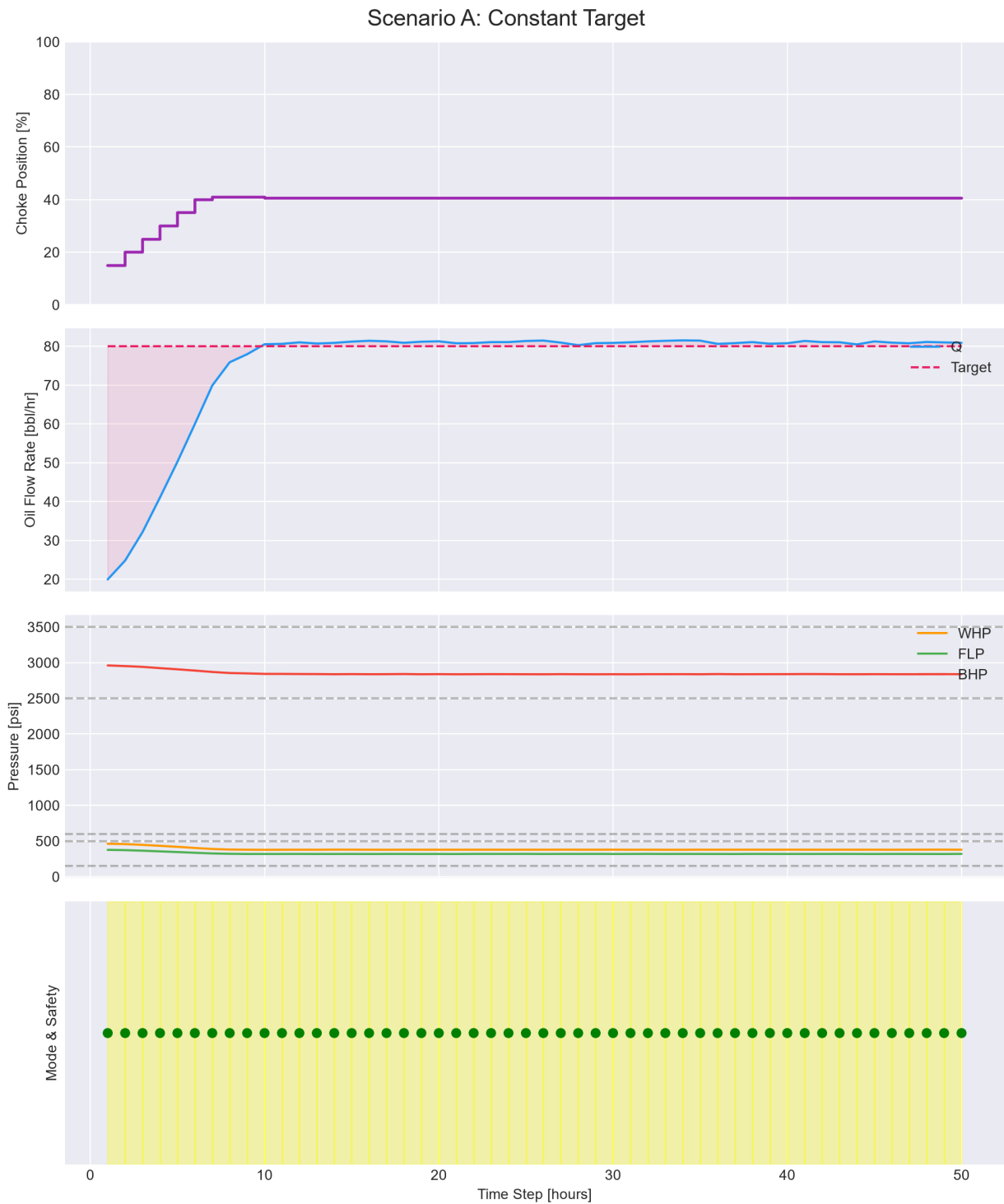


Figure 1: Scenario A: 6-panel performance display. Choke settled at 40.5%, $Q=80.9$ bbl/hr, 0 violations.

Scenario B: Target Change (60 -> 120 bbl/hr)

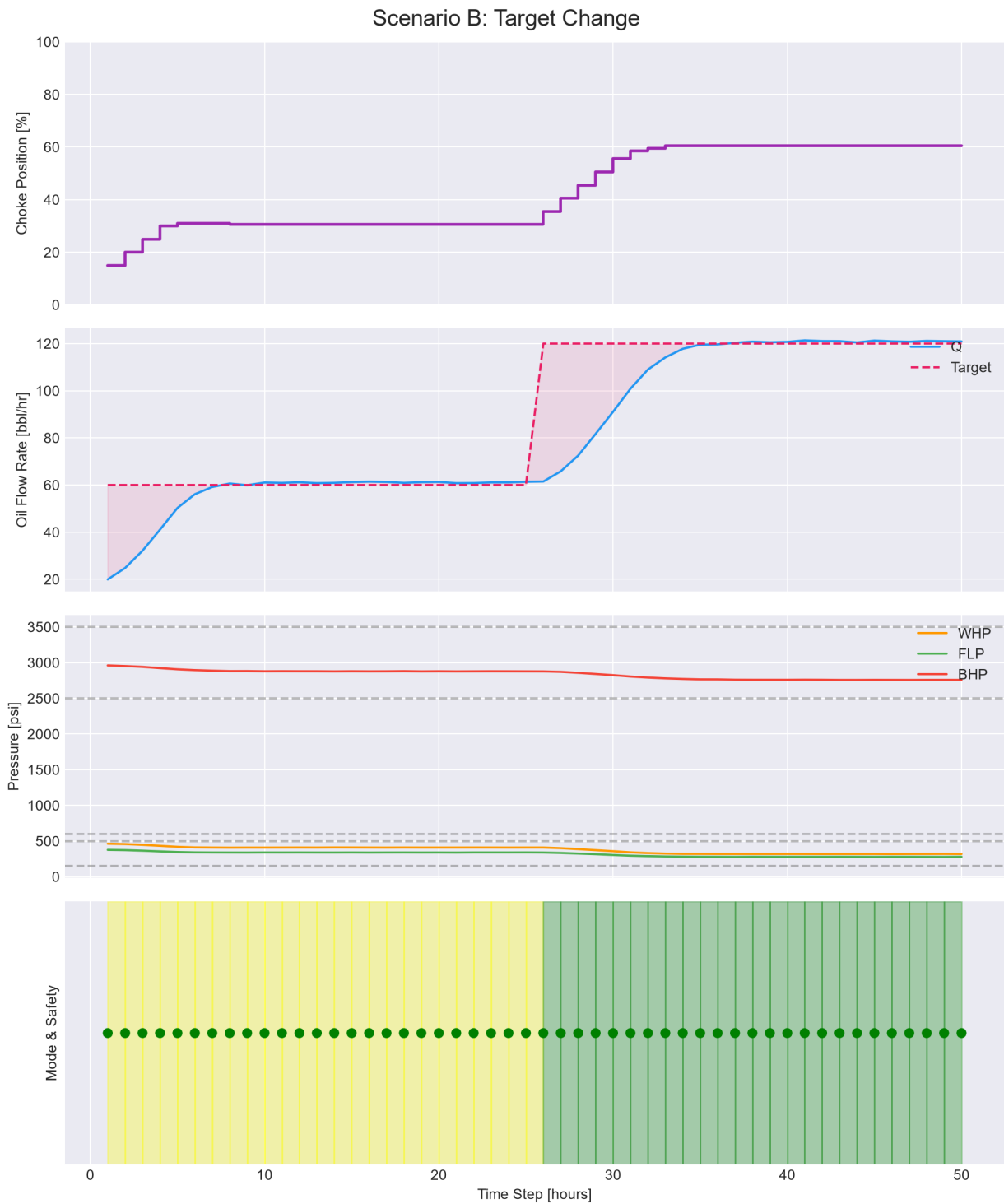


Figure 2: Scenario B: Setpoint transition. Smooth ramp, TRACKING mode, 0 violations.

Scenario C: Infeasible Target ($Q=250$ bbl/hr)

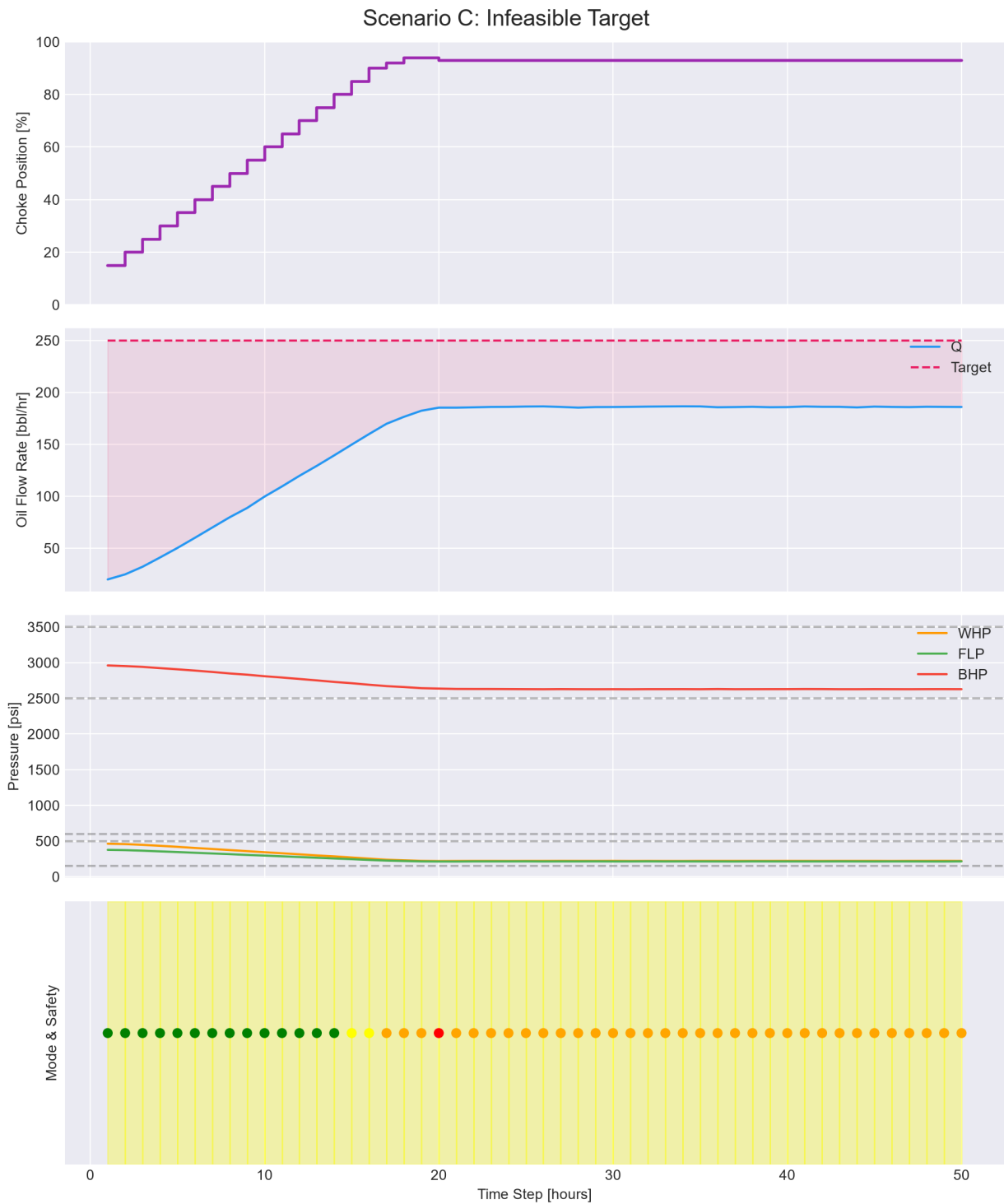


Figure 3: Scenario C: Constraint-limited max production. Choke at 93%, Q=186 bbl/hr.

Summary Dashboard

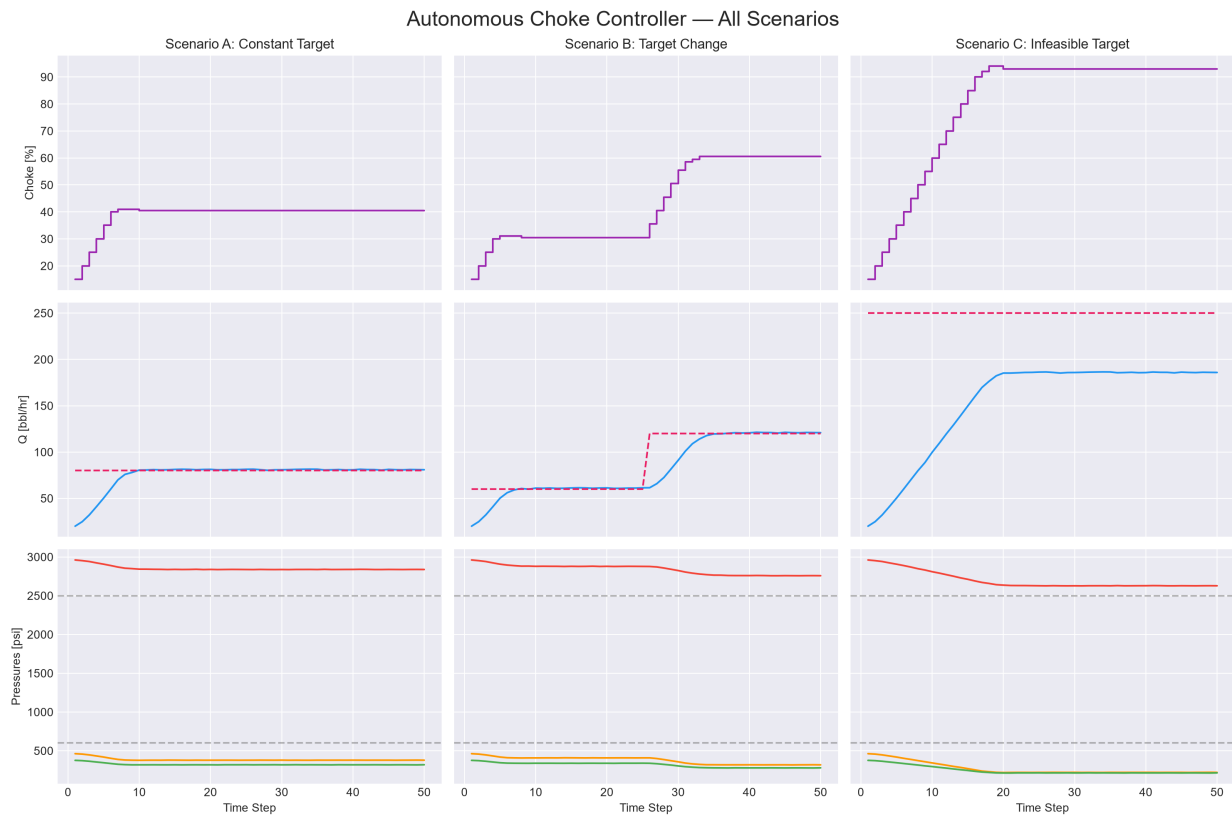


Figure 4: Summary dashboard: All 3 scenarios side-by-side comparison.

How to Regenerate

```
cd 01_Source_Code
python run_scenarios.py
# Outputs saved to output/ directory
```

All results are deterministic at seed=42. Running the pipeline produces identical (noise-seeded) outputs.